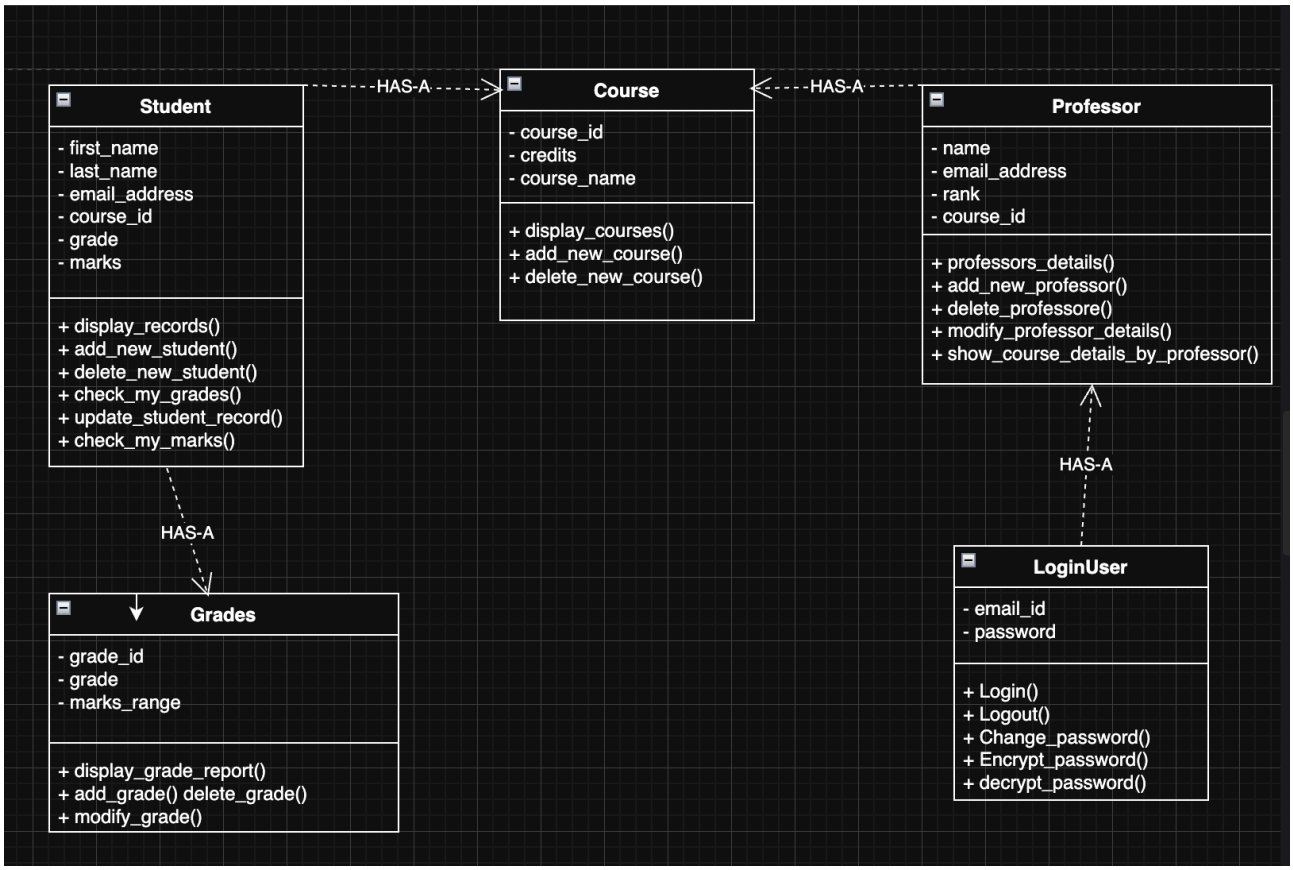
OOD:



Unit Test & Execution Times(In test_CheckMyGrade.py):

```
(base) wupeixuan@wupeixuandeMacBook-Air / % /usr/bin/python3 /Users/wupeixuan/Documents/GitHub/CheckMyGrade-app/test_CheckMyGrade.py
Course with this ID already exists.
.Professor already exists.
.Student already exists.
.New course added successfully.
Course deleted successfully.
.New professor added successfully.
Professor deleted successfully.
.New student added successfully.
Student deleted successfully.
.Course with this ID already exists.
.Professor already exists.
Professor record updated successfully.
.Search time: 0.0018339157104492188
.Sort time by email: 0.00010323524475097656
.Sort time by marks: 0.00011801719665527344
..Student already exists.
Student record updated successfully.
.
----------------------------------------------------------------------
Ran 13 tests in 0.034s

OK
```

Students

add_new_student: Adds a new student record after checking for duplicate email addresses.

```
print("\n=== Student Test ===")
student1 = Student("S0000", "Sam", "Carpenter", "sam@mycsu.edu", "DATA200", "A", 96)
```

```
=== Student Test ===
New student added successfully.
```

	A	B	C	D	E	F	G
993	**S0992**	Liam	Thompson	student0992@mycsu.edu	DATA208	A	98
994	**S0993**	Isabella	Miller	student0993@mycsu.edu	DATA209	A	94
995	**S0994**	Amelia	Lee	student0994@mycsu.edu	DATA200	C	73
996	**S0995**	Isabella	Lee	student0995@mycsu.edu	DATA206	F	57
997	**S0996**	Ava	Taylor	student0996@mycsu.edu	DATA207	C	71
998	**S0997**	Ava	Miller	student0997@mycsu.edu	DATA205	F	56
999	**S0998**	Ava	Thomas	student0998@mycsu.edu	DATA205	A	90
1000	**S0999**	Ava	Thompson	student0999@mycsu.edu	DATA203	F	54
1001	**S1000**	Liam	Kim	student1000@mycsu.edu	DATA207	F	56
1002	**S_TEST_00**	Test	User	testuser@mycsu.edu	DATA200	A	95
1003	**S_TEST_00**	Update	User	updateuser@mycsu.edu	DATA200	B	88
1004	**S0000**	Sam	Carpenter	sam@mycsu.edu	DATA200	A	96

check_my_grades & check_my_score: Displays the grades and marks of a specific student.

```
student1 = Student("Sam", "Carpenter", "sam@mycsu.edu", "DATA200", "A", 96)

'''student1.add_new_student()
student1.display_records()'''
student1.check_my_grades()
student1.check_my_marks()
```

```
=== Student Test ===
Sam Carpenter's grades: A
Sam Carpenter's marks: 96
```

update_student_record: Updates an existing student's information such as name, course, grades, or marks.

```
print("\n=== Update Student Test ===")
student1.update_student_record(new_marks=92, new_grades="A")
```

```
(base) wupeixuan@wupeixuandeMacBook-Air / % /usr/bin/python3 /Users/wupeixuan/Documents/GitHub/CheckMyGrade-app/CheckMyGrade.py

=== Student Test ===

=== Update Student Test ===
Student record updated successfully.
```

	A	B	C	D	E	F	G
995	S0994	Amelia	Lee	student0994@mycsu.edu	DATA200	C	73
996	S0995	Isabella	Lee	student0995@mycsu.edu	DATA206	F	57
997	S0996	Ava	Taylor	student0996@mycsu.edu	DATA207	C	71
998	S0997	Ava	Miller	student0997@mycsu.edu	DATA205	F	56
999	S0998	Ava	Thomas	student0998@mycsu.edu	DATA205	A	90
1000	S0999	Ava	Thompson	student0999@mycsu.edu	DATA203	F	54
1001	S1000	Liam	Kim	student1000@mycsu.edu	DATA207	F	56
1002	S_TEST_00	Test	User	testuser@mycsu.edu	DATA200	A	95
1003	S_TEST_00	Update	User	updateuser@mycsu.edu	DATA200	B	88
1004	S0000	Sam	Carpenter	sam@mycsu.edu	DATA200	A	92

search_student: Searches for a student using their student id and displays the student's record. It also calculates and prints the time taken to perform the search.

```
print("\n=== Search Student Test ===")
Student.search_student("S0001")
```

```
=== Search Student Test ===
Student Record
Student ID: S0001
Name: Ava Lee
Email: student0001@mycsu.edu
Course ID: DATA206
Grades: D
Marks: 65
Search time: 0.000015 seconds
```

sort_student_by_name: Sorts student records by name in ascending order.

```
Student Record
Student ID: S0957
Name: Oliver Lee
Email: student0957@mycsu.edu
Course ID: DATA201
Grades: B
Marks: 83
--------------------
Student Record
Student ID: S0974
Name: Oliver Perez
Email: student0974@mycsu.edu
Course ID: DATA206
Grades: C
Marks: 77
--------------------
Student Record
Student ID: S0989
Name: Oliver Thompson
Email: student0989@mycsu.edu
Course ID: DATA203
Grades: D
Marks: 67
--------------------
Student Record
Student ID: S0000
Name: Sam Carpenter
Email: sam@mycsu.edu
Course ID: DATA200
Grades: A
Marks: 92
--------------------
Student Record
Student ID: S_TEST_001
Name: Test User
Email: testuser@mycsu.edu
Course ID: DATA200
Grades: A
Marks: 95
--------------------
Student Record
Student ID: S_TEST_002
Name: Update User
Email: updateuser@mycsu.edu
Course ID: DATA200
Grades: B
Marks: 88
```

sort_student_by_marks: Sorts student records by marks in ascending order.

```
Student Record
Student ID: S0971
Name: James Miller
Email: student0971@mycsu.edu
Course ID: DATA207
Grades: A
Marks: 99
--------------------
Student Record
Student ID: S0984
Name: Isabella Chen
Email: student0984@mycsu.edu
Course ID: DATA204
Grades: A
Marks: 99
--------------------
Student Record
Student ID: S0051
Name: James Garcia
Email: student0051@mycsu.edu
Course ID: DATA208
Grades: A
Marks: 100
--------------------
Student Record
Student ID: S0057
Name: James Chen
Email: student0057@mycsu.edu
Course ID: DATA207
Grades: A
Marks: 100
--------------------
Student Record
Student ID: S0063
Name: Noah Chen
Email: student0063@mycsu.edu
Course ID: DATA206
Grades: A
Marks: 100
--------------------
Student Record
Student ID: S0186
Name: William Kim
Email: student0186@mycsu.edu
Course ID: DATA209
Grades: A
Marks: 100
--------------------
Student Record
Student ID: S0194
```

Calculates and displays the average marks and median for a specific course.

```
=== Average by Course ===
Average marks for DATA200: 74.06451612903226

=== Median by Course ===
Median marks for DATA200: 74
```

generate_report_by_course: Displays all student records and grades for a specific course.

```
=== Report by Course ===
Grade Report for Course: DATA200
S0039 | Emma Kim | F | 58
S0065 | James Lee | D | 65
S0069 | James Taylor | D | 67
S0093 | Emma Garcia | B | 85
S0095 | Liam Kim | C | 78
S0098 | Emma Lee | F | 50
S0105 | Oliver Thomas | D | 65
S0113 | Liam Taylor | D | 60
S0114 | Oliver Lee | B | 84
S0134 | Benjamin Kim | D | 64
S0136 | Oliver Perez | D | 66
S0152 | James Miller | F | 53
S0158 | Amelia Wang | A | 97
S0172 | Ava Perez | B | 84
S0178 | William Thompson | F | 57
S0188 | James Taylor | D | 62
S0227 | Ava Kim | D | 63
S0247 | Noah Perez | F | 54
S0256 | James Kim | C | 74
S0272 | Amelia Chen | B | 83
S0273 | Noah Chen | F | 51
S0281 | Isabella Chen | D | 64
```

generate_report_by_student: Displays all student records and grades for a specific student.

```
=== Report by Student ===
Grade Report for Student
Student Record
Student ID: S0000
Name: Sam Carpenter
Email: sam@mycsu.edu
Course ID: DATA200
Grades: A
Marks: 92
```

delete_new_student: Deletes a student record based on the provided email address.

```
========== Delete Student ==========
Student deleted successfully.
```

998	**S0997**	Ava	Miller	student0997@mycsu.edu	DATA205	F	56
999	**S0998**	Ava	Thomas	student0998@mycsu.edu	DATA205	A	90
1000	**S0999**	Ava	Thompson	student0999@mycsu.edu	DATA203	F	54
1001	**S1000**	Liam	Kim	student1000@mycsu.edu	DATA207	F	56
1002	**S_TEST_00**	Test	User	testuser@mycsu.edu	DATA200	A	95
1003	**S_TEST_00**	Update	User	updateuser@mycsu.edu	DATA200	B	88

Course

```python
# Course Demo
print("\n=== Course Test ===")
course1 = Course("DATA200", "Data Science", "Provides insight about DS and Python")
course1.add_new_course()
course1.display_courses()
'''print("\n=== Delete Course===")
Course.delete_new_course("DATA200")'''
```

add_new_course: Adds a new course if the course ID does not already exist.

```
=== Course Test ===
New course added successfully.
```

	course_id	course_name	description
2	**DATA201**	Python Programming	Covers Python basics and object-oriented programming
3	**DATA202**	Statistics	Introduces descriptive and inferential statistics
4	**DATA203**	Machine Learning	Explains supervised and unsupervised learning methods
5	**DATA204**	Database Systems	Covers relational databases and SQL concepts
6	**DATA205**	Data Visualization	Focuses on visual representation of data
7	**DATA206**	Big Data	Introduces large-scale data processing concepts
8	**DATA207**	Cloud Computing	Explains cloud platforms and deployment models
9	**DATA208**	Data Mining	Covers pattern discovery and data analysis techniques
10	**DATA209**	Algorithms	Introduces algorithm design and problem-solving methods
11	**TEST101**	Test Course	Test Description
12	**MOD101**	New Course	New Description
13	**DATA200**	Data Science	Provides insight about DS and Python

display_courses: Displays course details including course ID, name, and description.

```
Course Record
Course ID: DATA200
Course Name: Data Science
Description: Provides insight about DS and Python
```

delete_new_course: Deletes a course based on the course ID.

```
=== Delete Course===
Course deleted successfully.
```

1	course_id	course_name	description
2	**DATA201**	Python Programming	Covers Python basics and object-oriented programming
3	**DATA202**	Statistics	Introduces descriptive and inferential statistics
4	**DATA203**	Machine Learning	Explains supervised and unsupervised learning methods
5	**DATA204**	Database Systems	Covers relational databases and SQL concepts
6	**DATA205**	Data Visualization	Focuses on visual representation of data
7	**DATA206**	Big Data	Introduces large-scale data processing concepts
8	**DATA207**	Cloud Computing	Explains cloud platforms and deployment models
9	**DATA208**	Data Mining	Covers pattern discovery and data analysis techniques
10	**DATA209**	Algorithms	Introduces algorithm design and problem-solving methods
11	**TEST101**	Test Course	Test Description
12	**MOD101**	New Course	New Description

Professor

```python
# Professor Demo
print("\n=== Professor Test ===")
professor1 = Professor(
    "michael.johnson@mycsu.edu",
    "Michael Johnson",
    "Senior Professor",
    "DATA200"
)

professor1.add_new_professor()
professor1.professors_details()
```

add_new_professor: Adds a new professor if the professor ID is unique.

```
=== Professor Test ===
New professor added successfully.
```

professors

	professor_id	professor_name	rank	course_id
1				
2	sarah.lee@mycsu.edu	Sarah Lee	Assistant Professor	DATA201
3	david.wang@mycsu.edu	David Wang	Senior Professor	DATA202
4	emily.chen@mycsu.edu	Emily Chen	Associate Professor	DATA203
5	robert.brown@mycsu.edu	Robert Brown	Senior Professor	DATA204
6	linda.garcia@mycsu.edu	Linda Garcia	Associate Professor	DATA205
7	james.wilson@mycsu.edu	James Wilson	Senior Professor	DATA206
8	sophia.martinez@mycsu.edu	Sophia Martinez	Senior Professor	DATA207
9	daniel.kim@mycsu.edu	Daniel Kim	Assistant Professor	DATA208
10	olivia.clark@mycsu.edu	Olivia Clark	Associate Professor	DATA209
11	test.prof@mycsu.edu	Test Professor	Assistant Professor	DATA200
12	modify.prof@mycsu.edu	New Professor	Senior Professor	DATA200
13	michael.johnson@mycsu.edu	Michael Johnson	Senior Professor	DATA200

professors_details: Displays the details of a professor including ID, name, rank, and course.

```
Professor Record
Professor ID: michael.johnson@mycsu.edu
Professor Name: Michael Johnson
Rank: Senior Professor
Course ID: DATA200
```

show_course_details_by_professor: Displays the course details taught by a specific professor.

```
=== Professor Test ===
Course taught by this professor:
Course Record
Course ID: DATA200
Course Name: Data Science
Description: Provides insight about DS and Python
```

modify_professor_details: Updates a professor's details such as name, rank, or course.

```
print("\n=== Modify Professor Test ===")
professor1.modify_professor_details(new_professor_name="Michael J. Johnson")
```

```
=== Modify Professor Test ===
Professor record updated successfully.
```

10	olivia.clark@mycsu.edu	Olivia Clark	Associate Professor	DATA209
11	test.prof@mycsu.edu	Test Professor	Assistant Professor	DATA200
12	modify.prof@mycsu.edu	New Professor	Senior Professor	DATA200
13	michael.johnson@mycsu.edu	Michael J. Johnson	Senior Professor	DATA200

generate_report_by_professor: Displays all student records for the course taught by a specific professor.

```
=== Report by Professor ===
Grade Report for Professor: Michael J. Johnson
Course ID: DATA200
Amelia Thompson | D | 64
Noah Davis | A | 99
Charlotte Williams | B | 82
Harper Brown | C | 78
James Thomas | F | 53
Alexander Miller | D | 62
Amelia Garcia | B | 82
William Lee | A | 94
Evelyn Martinez | D | 61
William Smith | D | 62
Amelia Moore | A | 94
Ava Perez | C | 76
Abigail Lee | F | 58
Evelyn Perez | C | 75
Oliver Jackson | D | 66
Oliver Johnson | D | 64
Elijah Perez | D | 69
Ava Wilson | A | 92
Ava Martin | F | 57
Charlotte Moore | B | 89
Amelia Martinez | A | 100
Lucas Taylor | C | 74
William Jackson | A | 92
Charlotte Lee | A | 90
Alexander Smith | A | 94
Ava Martinez | F | 51
William Brown | A | 93
William Johnson | F | 57
Abigail Smith | A | 91
Amelia Lopez | B | 81
Ava Thompson | D | 64
Noah Taylor | A | 97
```

delete_professor: Deletes a professor based on the professor ID.

```
=== Delete Professor Test ===
Professor deleted successfully.
```

professors

	professor_id	professor_name	rank	course_id
2	sarah.lee@mycsu.edu	Sarah Lee	Assistant Professor	DATA201
3	david.wang@mycsu.edu	David Wang	Senior Professor	DATA202
4	emily.chen@mycsu.edu	Emily Chen	Associate Professor	DATA203
5	robert.brown@mycsu.edu	Robert Brown	Senior Professor	DATA204
6	linda.garcia@mycsu.edu	Linda Garcia	Associate Professor	DATA205
7	james.wilson@mycsu.edu	James Wilson	Senior Professor	DATA206
8	sophia.martinez@mycsu.edu	Sophia Martinez	Senior Professor	DATA207
9	daniel.kim@mycsu.edu	Daniel Kim	Assistant Professor	DATA208
10	olivia.clark@mycsu.edu	Olivia Clark	Associate Professor	DATA209
11	test.prof@mycsu.edu	Test Professor	Assistant Professor	DATA200
12	modify.prof@mycsu.edu	New Professor	Senior Professor	DATA200

Grade

add_grade: Adds a new grade record if the grade ID is unique.

```
=== Grade Test ===
New grade added successfully.
```

grades

grade_id	grade	marks_range
G001	A	90-100

display_grade_report: Displays grade information including grade and marks range.

```
Grade Record
Grade ID: G001
Grade: A
Marks Range: 90-100
```

modify_grade: Updates grade details such as grade value or marks range.

```
print("\n=== Modify Grade Test ===")
grade1.modify_grade(new_grade="A+", new_marks_range="95-100")
```

```
=== Modify Grade Test ===
Grade record updated successfully.
```

grades

	grade_id	grade	marks_range
1			
2	G001	A+	95-100

delete_grade: Displays grade information including grade and marks range.

```
=== Delete Grade ===
Grade deleted successfully.
```

grades

	grade_id	grade	marks_range
1			

LoginUser

add_user: Adds a new user to the system with an encrypted password.

```
=== Login Test ===
User added successfully.
```

login

	email_id	password
1	email_id	password
2	michael.johnson@mycsu.edu	134d15b0ce9f4a9f6f224a2d64cd6c7edf090146ed5fa503c6615a61fa8d853a
3	sarah.smith@mycsu.edu	6d0a87d84f373ae487420a92d3b8257b2ffc560f963db5ad626b7d5712eb49db
4	david.brown@mycsu.edu	40299c92585f90e0084690ce54936b90a228cb98efb6ad541fbba9cb664d785c
5	emily.davis@mycsu.edu	93ec601e81a69c061d01b469a0f589d719af49ac831e3774aba7b225d266492d
6	john.miller@mycsu.edu	ac4144952e8b9aa09839a6cb954aa72e52ff70f68598c77315a67ab0257bcefd
7	sophia.wilson@mycsu.edu	98fa57141899180452fe5110ab8a193a44c5bec194604c480c145d802e8916ac
8	daniel.moore@mycsu.edu	482187a930813175ed1b83d8e7669bc1214ee553ee7dbcccbdffc0811caf3ba1
9	olivia.taylor@mycsu.edu	3ec4dd9d812aebcac4bf13adcd9decba8d8bfb1bd53c3af2d19d015ae30a7950
10	james.anderson@mycsu.edu	5b1220dc81423eb8a1d7ef524b96119d4ffc06c73d3bff65c3038f38cd55f7b5
11	emma.thomas@mycsu.edu	7e45b6059eef4f6f94765633cfa1dbd8420255341b647df59b882e23c04c8fb1
12	johnson@mycsu.edu	84f26931a905f62a199330a5b683df18951715974cd2a8efc93de5f000aa07c0

login: Verifies user credentials by comparing the encrypted password.

```
Login successful.
```

logout: Logs out the current user from the system.

```
Logout successful.
```

change_password: Updates the user's password and stores it in encrypted form.

```
print("\n=== Change Password Test ===")
user1.change_password("NewPassword123#_")
```

```
=== Change Password Test ===
Password changed successfully.
```

login

	email_id	password
1	michael.johnson@mycsu.edu	134d15b0ce9f4a9f6f224a2d64cd6c7edf090146ed5fa503c6615a61fa8d853a
2	sarah.smith@mycsu.edu	6d0a87d84f373ae487420a92d3b8257b2ffc560f963db5ad626b7d5712eb49db
3	david.brown@mycsu.edu	40299c92585f90e0084690ce54936b90a228cb98efb6ad541fbba9cb664d785c
4	emily.davis@mycsu.edu	93ec601e81a69c061d01b469a0f589d719af49ac831e3774aba7b225d266492d
5	john.miller@mycsu.edu	ac4144952e8b9aa09839a6cb954aa72e52ff70f68598c77315a67ab0257bcefd
6	sophia.wilson@mycsu.edu	98fa57141899180452fe5110ab8a193a44c5bec194604c480c145d802e8916ac
7	daniel.moore@mycsu.edu	482187a930813175ed1b83d8e7669bc1214ee553ee7dbcccbdffc0811caf3ba1
8	olivia.taylor@mycsu.edu	3ec4dd9d812aebcac4bf13adcd9decba8d8bfb1bd53c3af2d19d015ae30a7950
9	james.anderson@mycsu.edu	5b1220dc81423eb8a1d7ef524b96119d4ffc06c73d3bff65c3038f38cd55f7b5
10	emma.thomas@mycsu.edu	7e45b6059eef4f6f94765633cfa1dbd8420255341b647df59b882e23c04c8fb1
11	johnson@mycsu.edu	f2160a384f868119db0bddcc8c98a151e3d4189f63365b5118690fde2df71fc6